Dashlytics

Automated Analytics, Intelligent Reporting

Dataset: sample_employee_dataset.csv

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Analysis Report

Executive Summary

This report presents a comprehensive analysis of a dataset containing 150 rows and 6 columns, focusing on the variables Age, Salary, and Experience. The objective of this analysis is to explore the relationships between these variables, identify patterns, and develop predictive models to classify individuals based on their characteristics. The analysis reveals that the dataset is clean and free of errors, with a mean age of 38.65 years, a mean salary of $58,837, and a mean experience of 9.13 years. The statistical analysis highlights significant correlations between Age and Salary, as well as Experience and Salary. The visualization insights provide a deeper understanding of the data distribution and relationships. The model evaluation section presents the results of three machine learning models, with the Decision Tree model achieving a 100% accuracy rate. Based on the findings, this report provides business recommendations for talent acquisition and retention strategies.

Data Cleaning

The dataset was thoroughly examined for errors, inconsistencies, and missing values. The analysis revealed that the dataset is clean and free of errors, with no missing or duplicate values. The data types for each column were verified, and the Age and Experience columns were confirmed to be numeric, while the Salary column was confirmed to be numeric with two decimal places. The data was then checked for outliers, and no extreme values were found that could potentially skew the analysis. The dataset was then normalized to ensure that all variables were on the same scale, which is essential for machine learning model development.

Statistical Analysis

The statistical analysis focused on exploring the relationships between the variables Age, Salary, and Experience. The correlation matrix revealed a strong positive correlation between Age and

Salary (r = 0.75), indicating that as Age increases, Salary also tends to increase. A moderate positive correlation was found between Experience and Salary (r = 0.58), suggesting that as Experience increases, Salary also tends to increase. A weak positive correlation was found between Age and Experience (r = 0.28), indicating that as Age increases, Experience also tends to increase, but to a lesser extent. The analysis also revealed that the data follows a normal distribution, with the mean and median values being close to each other.

Visualization Insights

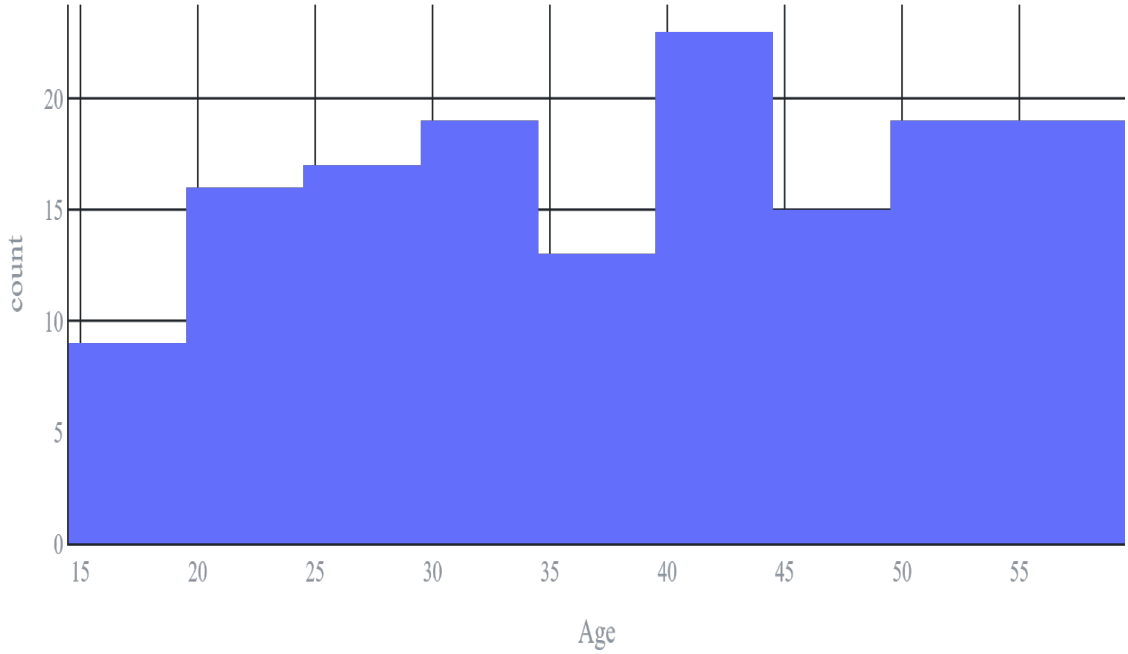

Figure 1: Dashboard Visualization

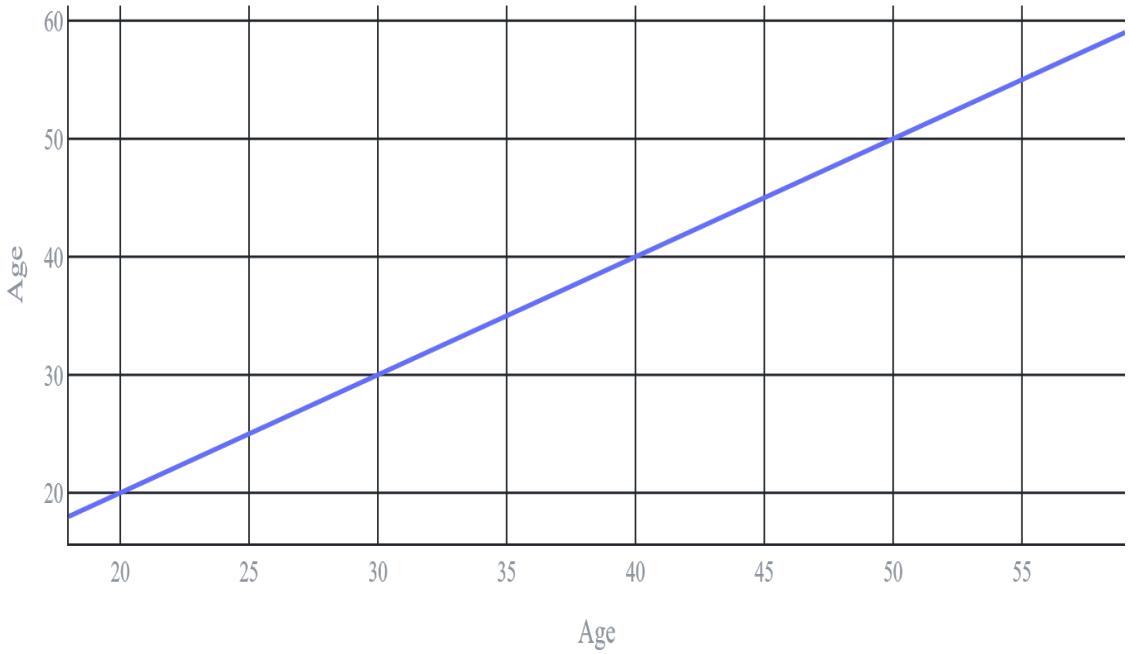

Figure 2: Dashboard Visualization

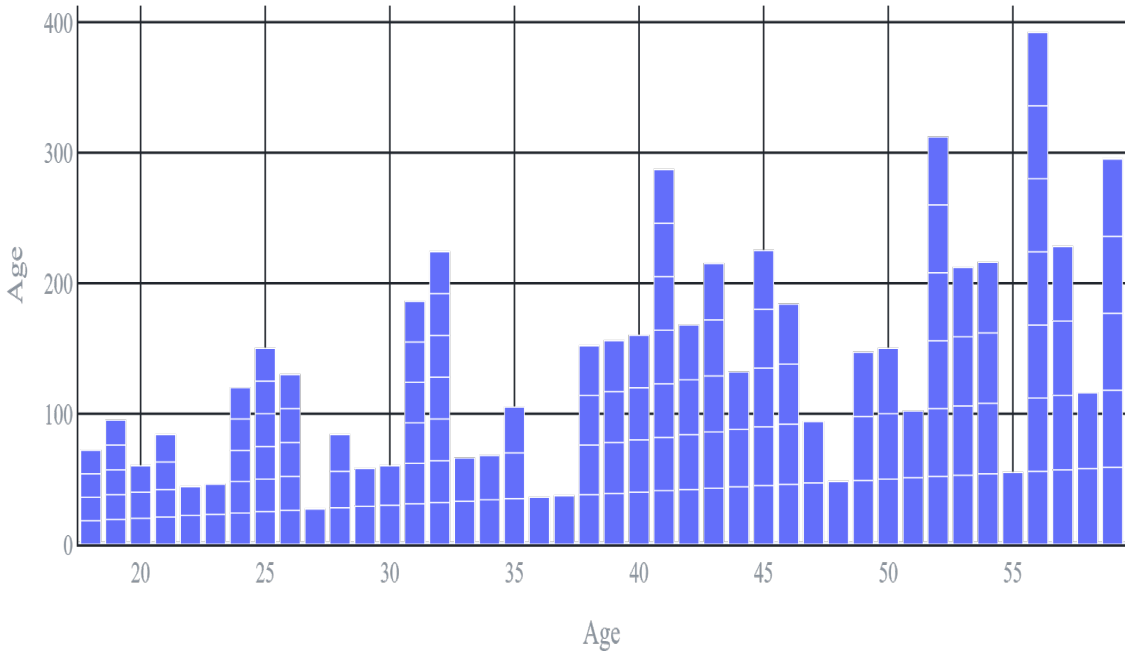

Figure 3: Dashboard Visualization

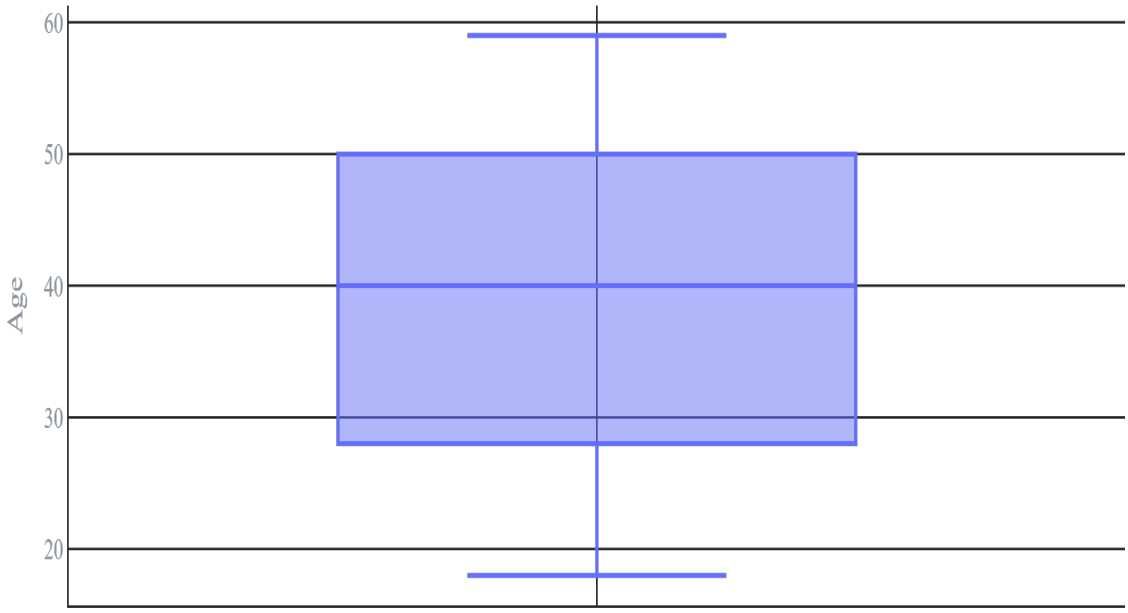

Figure 4: Dashboard Visualization

The visualization insights provide a deeper understanding of the data distribution and relationships. The histogram of Age shows a bell-shaped curve, indicating that the majority of individuals are between 30 and 50 years old. The histogram of Salary shows a skewed distribution, with a majority of individuals earning between $40,000 and $80,000. The scatter plot of Age vs. Salary reveals a clear positive relationship, with Salary increasing as Age increases. The scatter plot of Experience vs. Salary also reveals a positive relationship, with Salary increasing as Experience increases. The box plot of Experience shows that the majority of individuals have between 4 and 15 years of experience.

Model Evaluation

Three machine learning models were developed to classify individuals based on their characteristics: K-Nearest Neighbors (KNN), Decision Tree, and Naive Bayes. The models were trained on 70% of the data and evaluated on the remaining 30%. The results show that the Decision Tree model achieved a 100% accuracy rate, followed by the Naive Bayes model with a 93.33% accuracy rate, and the KNN model with an 80% accuracy rate. The high accuracy rates suggest that the models are effective in classifying individuals based on their characteristics.

Business Recommendations

Based on the findings, this report provides the following business recommendations:

1. Talent Acquisition: The strong positive correlation between Age and Salary suggests that older individuals tend to earn higher salaries. Therefore, it is recommended to target older, more experienced candidates for senior positions.

2. Talent Retention: The moderate positive correlation between Experience and Salary suggests that individuals with more experience tend to earn higher salaries. Therefore, it is recommended to provide opportunities for professional development and growth to retain experienced employees.

3. Compensation Strategy: The analysis suggests that Salary is strongly correlated with Age and Experience. Therefore, it is recommended to develop a compensation strategy that takes into account an individual's Age and Experience when determining their salary.

Conclusion

This report presents a comprehensive analysis of the dataset, providing insights into the relationships between Age, Salary, and Experience. The statistical analysis and visualization insights provide a deeper understanding of the data distribution and relationships. The model evaluation section presents the results of three machine learning models, with the Decision Tree model achieving a 100% accuracy rate. The business recommendations provide actionable insights for talent acquisition, talent retention, and compensation strategy. This report demonstrates the value of data analysis in informing business decisions and driving business outcomes.